

Web-Based Topup Cards & Reload Distribution System for Linaya Distributors

T. T. B. De Silva

Index Number – 1602799

Supervisor – Mr. M.P.Sarath Wijeshinghe

Academic Year – 2020

**This Interim report is submitted in partial fulfillment of the requirement of the
Degree of Bachelor of Information Technology (external) of the
University Of Colombo School Of Computing**



Abstract

Linaya Distributors is the authorized recharge card distributor for the Kalutara area under Hutchison Lanka (Pvt) Ltd. They are distributing items such as recharge cards, reloads, and SIMs. Linaya has two sub-locations. Each sub-location has 5 sales representatives to distribute these items to the market.

Their normal practice is to record manually whatever they receive from the company (Hutch), Whatever they transfer to the sub-locations and the sales representatives. And also the sales of each sales representative are recorded manually in logbooks. There is no such way to request stocks for sub-locations from the main location. At the end of the day they want to know what their daily sales, target achievements are, the stock balance of each sub-location and the sales representative, and so on. All these things are more important to maintain a healthy stock, fulfill the market requirement. Most importantly the owner has to have a transparent view of his business. There is no way to track stock availability at the retailer level and retailers couldn't order stocks when they required. Through a web-based system will able to rectify all these issues.

To overcome these issues, time-consuming processes, and to increase their efficiency Linaya Distributor decided to go with a Sales and Distribution management system.

The system is web based and used server is Apache. It is based on client-server architecture and developed in accordance to object oriented principals therefore it can be easily extended in future. System is developed using languages such as PHP, HTML and Java Script. Unified Modeling Language (UML) is used for modeling. Database used is a MySQL database. Bootstrap framework has been used, therefore the system is accessible through several devices such as Laptops, Tablets and Mobile phones.

List of acronyms

PHP - Hypertext Preprocessor

SQL - Structured Query Language

AJAX - Asynchronous JavaScript and XML

CSS - Cascading Style Sheet

IT - Information Technology

URL - Uniform Resource Locator

DFD - Data Flow Diagram

UML - Unified Modelling Language

HTML - Hypertext Markup Language

HTTP - Hypertext Transfer Protocol

Table of Contents

1. Chapter 1 – Introduction.....	1
1.1 Introduction to the Top-up cards & Reload distribution of Linaya Distributors	1
1.2 The motivation for the Project.....	1
1.3 Objectives of the project	2
1.4 Scope of the project	2
1.5 Dissertation Overview	3
2. Chapter 2 – Analysis.....	4
2.1 Requirement gathering techniques.....	4
2.1.1 Interviews.....	4
2.1.2 On site observations	4
2.1.3 Forms and reports	4
2.1.4 Questionnaires	5
2.2 Analyzing the Current Manual System	5
2.3 Drawbacks of the Existing System.....	6
2.4 Requirement analysis	6
2.4.1 Functional requirements.....	6
2.4.2 Non-Functional requirements	7
2.5 Justification for the Selected Methodology	8
3. Chapter 3 – Design.....	9
3.1 Alternative solutions	9
3.2 Reasons for using the web based solution	9
3.3 Alternative Design Strategies.....	9
3.4 Comparison of Alternative Design Strategies.	10
3.5 Design diagrams	11
3.5.1 Use case diagram.....	11
3.5.2 Use case descriptions.....	12
3.5.3 Activity Diagram	14
3.5.4 Entity Relationship Diagram.....	15
3.6 Interface Design	15

Table of Figures

FIGURE 1 WATERFALL PROCESS MODEL	8
FIGURE 2 USE CASE DIAGRAM FOR THE PROPOSED SYSTEM	11
FIGURE 3 ACTIVITY DIAGRAM LOGIN	14
FIGURE 4 ACTIVITY DIAGRAM STOCK TRANSFER	14
FIGURE 5 ENTITY RELATIONSHIP DIAGRAM.....	15
FIGURE 6 LOGIN PAGE	15
FIGURE 7 ADMIN PANEL DASHBOARD	16
FIGURE 8 USER ACCOUNT CREATE	16

1. Chapter 1 – Introduction

1.1 Introduction to the Top-up cards & Reload distribution of Linaya Distributors

There are mainly two ways of recharging our mobile E-wallet. Those are by using a top-up card or through an E-reload. Customers who want to recharge their E-wallet, they can do this easily through a retailer shop or communication. Telecommunication operators use dealers for the distribution of recharge products. Dealers purchase these products and distribute those through their sales representatives to the retailers.

Linaya Distributors is the dealer, who distribute recharge products of Hutchison telecommunications in Kalutara District. The dealer is maintaining two sub-points to handle the distribution. Each sub-point has a demarcated area while sales representatives having allocated retailer shops.

1.2 The motivation for the Project

The final year project is an important and key element of the BIT degree program of the University of Colombo School of Computing which allows us to deepen our practical knowledge and learn how to use our skills.

Linaya Distributors, Kalutara is keeping manual records of their sales, purchases, stocks, and forecasting, etc. On the other hand, the dealer has no idea about the market requirement and stock level of the retailers. The owner of the dealer has seen the requirement for a system. He was interested in introducing a web-based distribution management system, which can electronically store sales, purchases, stock levels, market requirements, and generate various types of reports by a single click.

New System will increase the efficiency and accuracy of the above requirements and generate useful reports to analyze information that will be helpful to the management.

1.3 Objectives of the project

- Replacing the current manual system with an automated system.
- Reduce workload and improve the efficiency and productivity of the users.
- Keep track of all the activities collaboratively.
- Purchases history management.
- Market requirement management.
- Maintain sub-locations stock levels.
- Provide reports for administrative and research purposes.
- Quality maintenance and reduce errors in the records.
- Provide a user-friendly system.
- Practically use the knowledge and skills gained throughout this degree program.

1.4 Scope of the project

- Keep purchase and stock transfer records.
- Manage stock levels of sub-locations, sales representatives.
- Handle stock requests from sales representatives, and retailers.
- Update daily sales of each sales representative.
- Register new retailers and allocate them to a sales representative.
- Keep track of top-up cards.
- Analyze market requirements.
- Assign sales targets for sales representatives.
- Monitor target achievements.

1.5 Dissertation Overview

The main six chapters of the Dissertation including the Introduction chapter describes the outline flow of the software development. This provides a better understanding of the system.

Chapter 2 - Analysis

This chapter includes the description of the process of requirement identification and finalizing, requirements that have to be obtained in the new system such as user requirements, functional and non-functional requirements, techniques used for requirement gathering.

Chapter 3 - Design

The design chapter includes Database design, Interface, and System design. Use case diagrams, ER diagrams are used to further understanding of the system. The designing wireframe of the user interfaces is described up to some extent.

Chapter 4 - Implementation

Describes the techniques of the implementation process. Coding standards, Tools that are used, and the reason to choose them are explained with some examples.

Chapter 5 - Evaluation

Describe how the testing is done using some of the actual data and dummy data. Test cases, Test reports, and Error handling techniques are discussed with the provided samples.

Chapter 6 - Conclusion

This chapter summarizes the outcome of the system, how can future extend the system and achievements gained throughout the project.

2. Chapter 2 – Analysis

This is the process of analyzing user requirements for a new system. Understanding the problem and way to resolve it has to be done in this phase. Identifying the project goal and scope and finalizing the functional and non-functional requirements are the main objectives of this phase. This phase plays a major role in the system development because the understanding between the end-user and the system analyst build within this phase. If not, incomplete and incorrect specifications may cause an unsuccessful result.

2.1 Requirement gathering techniques

Requirement gathering is one of the most critical phases during system development. Productivity, quality, and the correctness of the system depend on how well we gathered the requirements. So it is important to use various types of requirement gathering techniques to identify the requirements.

2.1.1 Interviews

Most common fact-gathering technique. This system has many types of end-users in different levels. Interviewing the end-users were helpful to understand the distribution process and finalizing the requirements. Therefore, this is the main method used for fact gathering.

2.1.2 On site observations

System analyst watching how the staff carry out the entire process an how the activities get done through site visits is another fact-gathering technique. Sometimes the end-users unable to explain the requirement this method was helpful to clarify the requirement.

2.1.3 Forms and reports

Since the existing system was mainly a manual paper-based system and used spreadsheets, we can use those forms and reports to identify some facts and gained knowledge. So that might help to develop the system which is familiar to the end-user.

2.1.4 Questionnaires

When there is a large number of people, it's impractical to gather requirements by interviewing them individually. So, this technique is used to gather requirements from a large number of people.

2.2 Analyzing the Current Manual System

To understand the domain of the system, a domain analysis was carried out through certain fact gathering techniques. The current system has been divided into three logical subsystems as follows.

- Stock request & Stock Transfer
- Inventory Management
- Invoicing

Stock request & Stock Transfer

Required stocks for the sub location are requested from the main warehouse was done via an email. For their tracking they are maintain an excel workbook manually.

Inventory Management

For their reconciliation purposes they are updating an excel for stock receivables, issuance and returns from the DSRs. From that they are managing the DSR inventory and sub locations inventory. There is no tracking of which voucher card is transferred to the which DSR, but only the quantity.

Invoicing

There is a manual bill to the retailer when he purchases the vouchers.

2.3 Drawbacks of the Existing System

- Time consuming - It takes lot of time to view in historical data and take long time to enter and store the data.
- Manual inventory system is time consuming, as the business owner must keep track of inventory sales on a daily basis, while updating the excel manually at the end of the day.
- Invoice generation is time consuming, as all fields should fill manually and wasting unnecessary time to type each invoice manual.

2.4 Requirement analysis

2.4.1 Functional requirements

1. Order Management

This function helps to manage order requests, returns, and transfers. Accordingly, to the user levels order management get vary.

- Update stocks received to the main warehouse
- View stock request from DSRs & Retailers
- Transfer stocks to the sub-dealers & DSRs
- Invoice to the Retailer
- Accept stock returns from sub-dealers & DSRs
- Return stocks to main warehouse & sub-dealers
- Cancel order requests

2. Product Management

The system will maintain details of products

- Create new products
- Edit product details (E.g. Pricing)
- Delete or Expire products

3. Sub dealer Management

There are two sub-locations at the movement. By this, users can manage those locations individually.

- Create new sub-dealer points
- Check stock levels on each point

4. DSR Management

Dealer Sales Representatives, who are doing redistribution their products to the retailers called DSR.

- Allocate retailers
- Allocate products
- Monthly target setting
- Check stock levels each individual

5. Retailer Management

- Register new retailers
- Manage purchase history

2.4.2 Non-Functional requirements

1. Usability

The users of the system practiced to a manual paper-based system and some basic computer applications. So the system should be easy enough to use every one even who have not much technical knowledge.

2. Security

Log in to the system will be done using a user name and password.

Create user accounts and provide login details that will be done by a system admin. That will restrict unauthorized parties from accessing the system. The information in the system is confidential and critical.

3. Performance

The system should respond not more than a minimum of 10 seconds for every task performed by a user.

4. Availability

Also, when there is a system malfunctioning, the system should be available in 1 to 2 working days. So that the business process is not severely affected.

5. Reliability

The system based on the sales information and, the growth of the business depends on the system. So the 100% accuracy of the system is needed.

2.5 Justification for the Selected Methodology

Waterfall approach was first SDLC Model to be used widely in Software Engineering to ensure success of the project. In "The Waterfall" approach, the whole process of software development is divided into separate phases. In this Waterfall model, typically, the outcome of one phase acts as the input for the next phase sequentially.

The following illustration is a representation of the different phases of the Waterfall Model.

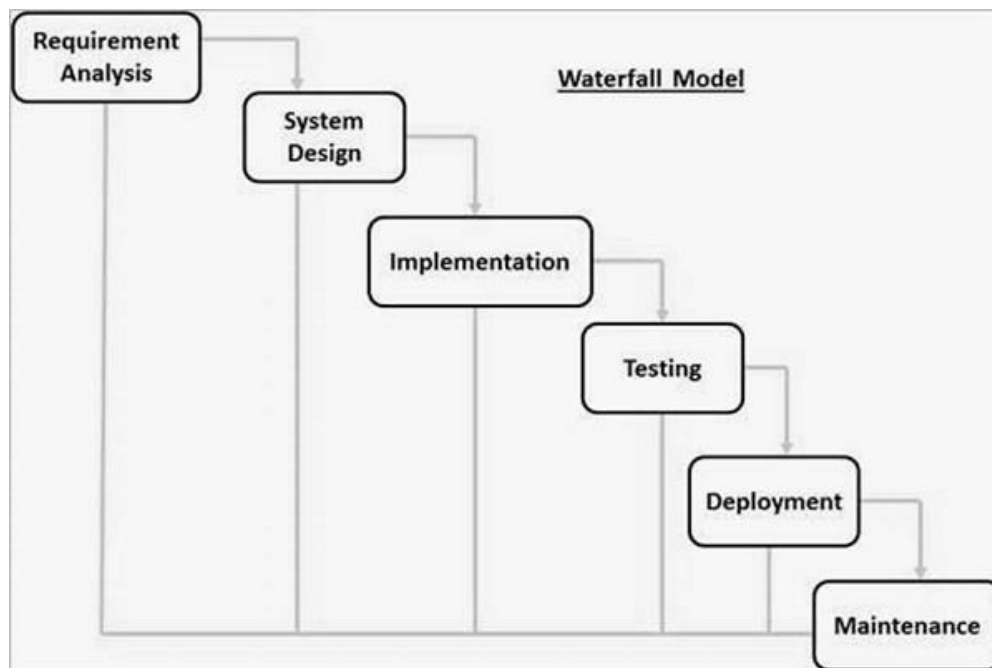


Figure 1 Waterfall process model

3. Chapter 3 – Design

In the design phase, will discuss the overall design and planning a solution for the problem identified in the requirement analysis. The quality of the final product highly depends on this phase.

3.1 Alternative solutions

- Develop a Standalone application.
- Web base Solution.

3.2 Reasons for using the web based solution

To meet client requirements, some users have to access the system through a mobile phone or a tablet. The telecommunication business highly depends on product availability. So, to manage the stock out situation in the market, their inventory has to get updated in real-time. Also, to get the retailer's request real-time best solution to go with the web-based system.

3.3 Alternative Design Strategies

Structured design is a conceptualization of problem into several well-organized elements of solution. It gives better understanding of how the problem is being solved. Structured design also makes it simpler for designer to concentrate on the problem more accurately. The small pieces of problem are solved by means of solution modules. A good structured design always follows some rules for communication among multiple modules,

- Cohesion -grouping of all functionally related elements.
- Coupling -communication between different modules.

A good structured design has high cohesion and low coupling arrangements.

Object Oriented Design works around the entities and their characteristics instead of functions involved in the software system. This design strategy focuses on entities and its characteristics. The whole concept of software solution revolves around the engaged entities.

Objects - All entities involved in the solution design are known as objects. Every entity has some attributes associated to it and has some methods to perform on the attributes.

Classes - A class is a generalized description of an object. An object is an instance of a class. Class defines all the attributes, which an object can have and methods, which defines the functionality of the object.

Encapsulation - In OOD, Encapsulation not only bundles important information of an object together, but also restricts access of the data and methods from the outside world. This is called information hiding.

Inheritance - OOD allows similar classes to stack up in hierarchical manner where the lower or sub-classes can import, implement and re-use allowed variables and methods from their immediate super classes. This property of OOD is known as inheritance. This makes it easier to define specific class and to create generalized classes from specific ones.

Polymorphism - OOD languages provide a mechanism where methods performing similar tasks but vary in arguments, can be assigned same name. This is called polymorphism, which allows a single interface performing tasks for different types. Depending upon how the function is invoked, respective portion of the code gets executed.

Abstraction - That hides the internal implementation details, just need to know which methods of the object are available to call and which input parameters are needed to trigger a specific operation.

3.4 Comparison of Alternative Design Strategies.

Structured Design	Object Oriented Design
Program Represent as a Logical Structure	Program written as a collection of objects which communicates with each other.
Use Sequential Methodology Ex: Waterfall Methodology	Uses Iterative and Incremental Methodologies. Ex: RUP,RAD
Low Code Reusability.	High Code Reusability.
Low Data Security.	High Data Security.
Uses Top Down Approach.	Uses Bottom Up Approach.

3.5 Design diagrams

3.5.1 Use case diagram

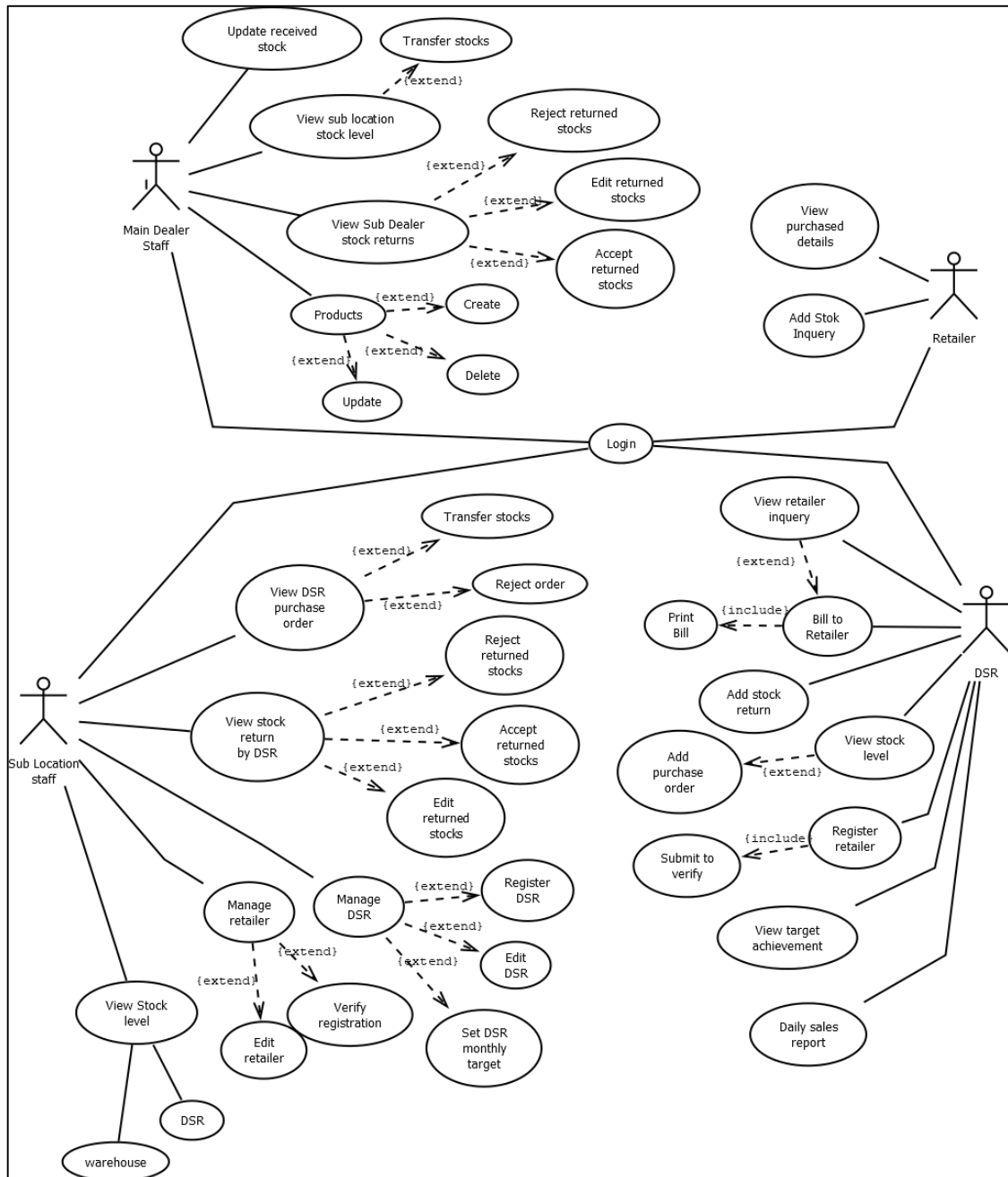


Figure 2 Use case diagram for the proposed system

3.5.2 Use case descriptions

Use Case description - Login

Use-case Number	UC-01	
Use-Case Name	Log in	
Priority	High	
Actor	Manager, Sub warehouse staff, DSR, Retailer	
Description	This use case describes how Staff members and Retailers login into the Linaya Distribution system	
Precondition	None	
Post-condition	If the use case was successful, the actor is now logged into the system. If not, the system state is unchanged.	
Basic course of Action	User Action	System Response
	1. The Actor is on the login page to login to the system. 3. The Actor enters username and password, click on a Login Button.	2. The system promotes the Actor to enter Username, Password. 4. The system verifies that all the filled have been filled out and valid. 5. The system successfully logged in the system. 6. Use case exit.
Alternate course of Action	5.1 If all fields are not filled out and not matched to the username and password the system notifies the actor a message check Username or Password and then goes back or returns to step 4 of basic course of Action to enter again.	

Table 1 Use case description - Login

Use Case description - Stock transfer DSR

Use-case Number	UC-02	
Use-Case Name	Stock transfer DSR	
Priority	High	
Actor	Sub warehouse staff	
Description	This use case describes how Staff member transfer stocks to a DSR	
Precondition	DSR Should request stocks by initiating a purchase order	
Post-condition	If the use case was successful, DSR's inventory get increased and sub inventory get reduced.	
Basic course of Action	User Action	System Response
	1. The Actor is in the stock transfer page. 3. The Actor assign batch number and the starting serial number range	2. The system promotes the Actor to assign batch number and the starting serial number range according to requested products. 4. The system verifies that all the products have been assign serials and check with the ordered quantity. 5. The system successfully transfer stocks to the relevant DSR and reduced stocks from sub warehouse inventory. 6. Use case exit.
Alternate course of Action	5.1 If all the products not assign with serial ranges system notify the actor to assign all the products. Also if there is no sufficient stock to transfer system will notify.	

Table 2 Use case description - Stock Transfer DSR

3.5.3 Activity Diagram

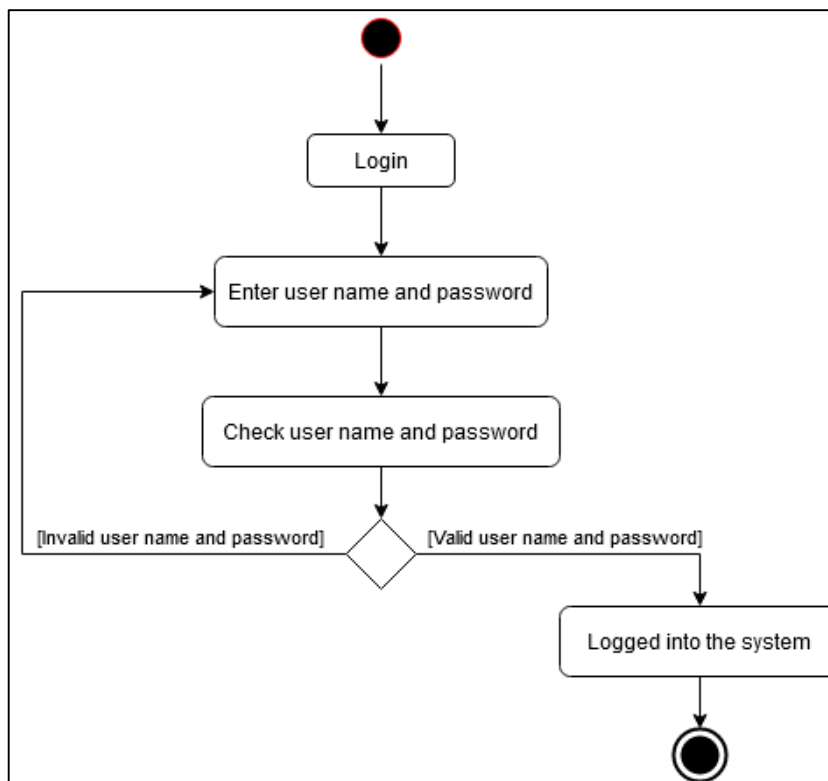


Figure 3 Activity Diagram Login

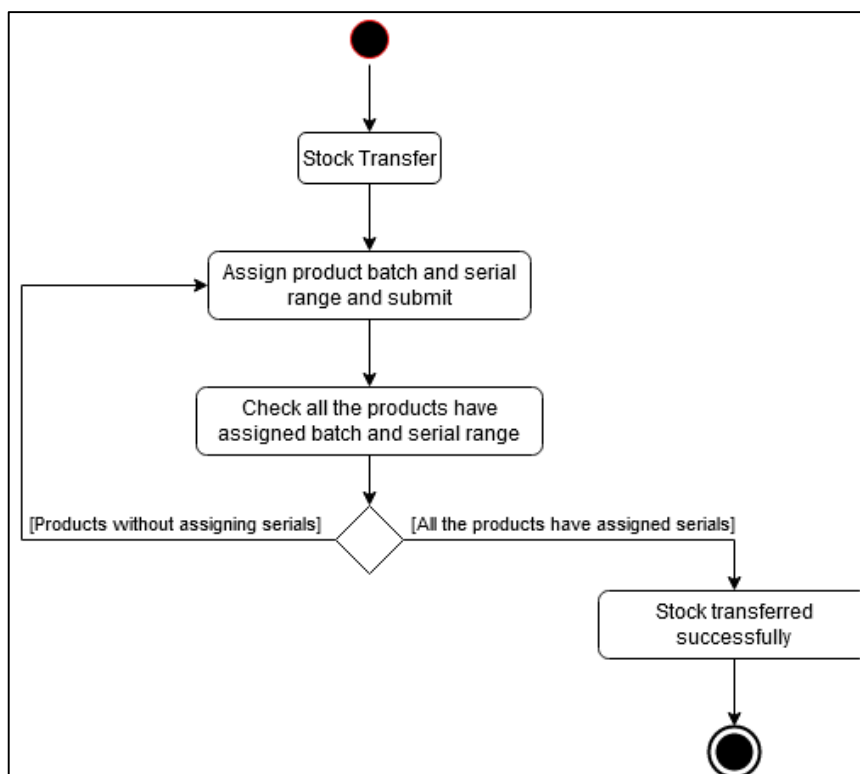


Figure 4 Activity Diagram Stock Transfer

3.5.4 Entity Relationship Diagram

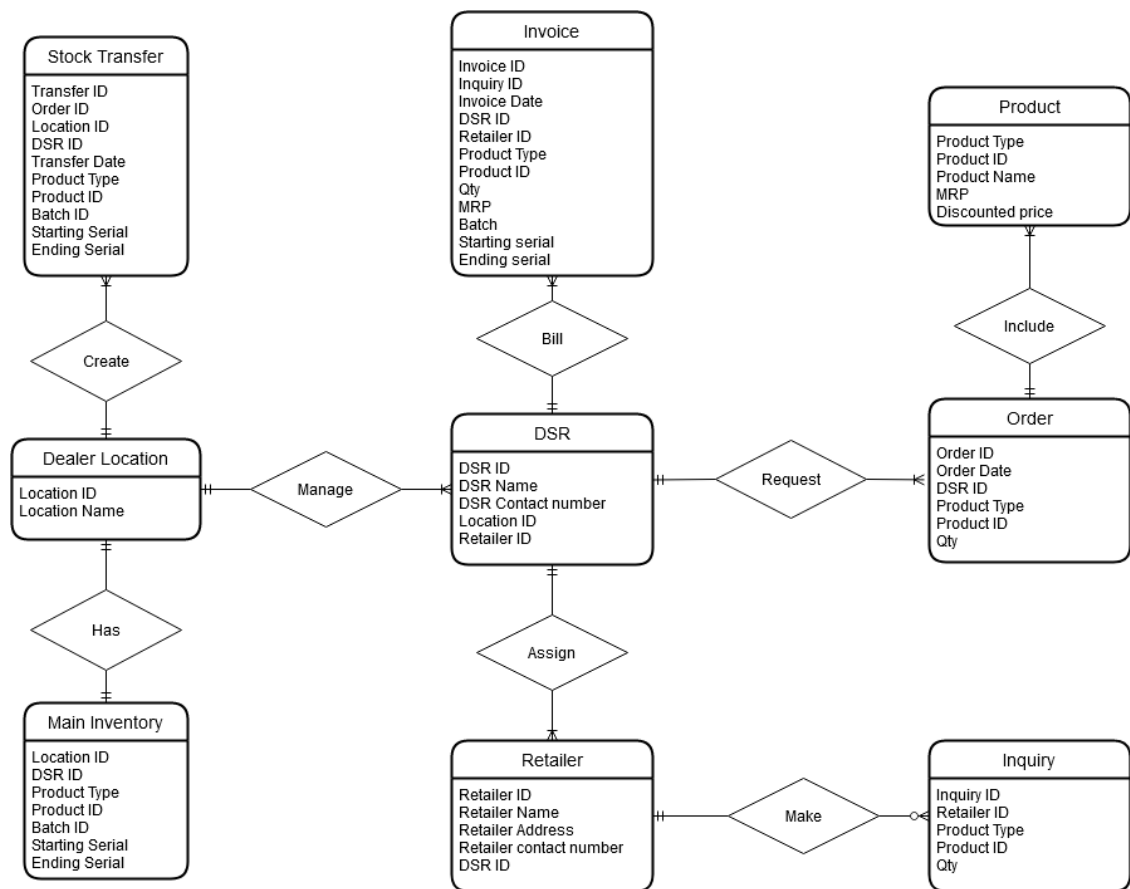


Figure 5 Entity Relationship Diagram

3.6 Interface Design

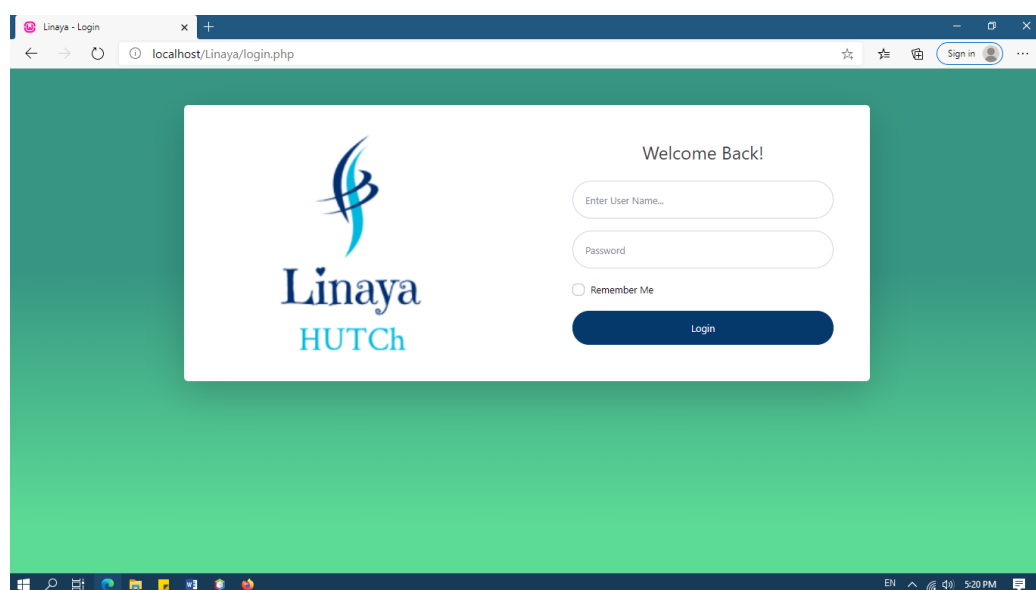


Figure 6 Login page

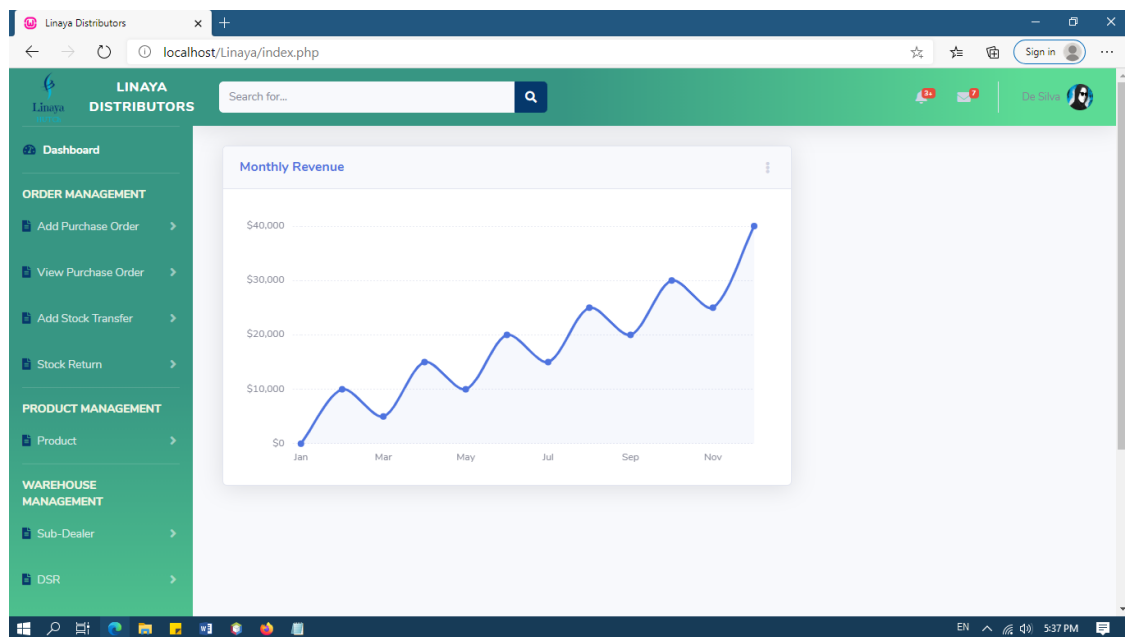


Figure 7 Admin panel Dashboard

Create User Account

Title: [Dropdown menu]

First Name: [Enter First Name]

Last Name: [Enter Last Name]

User Name: [Enter User Name]

Password: [Enter Password]

Password Confirm: [Re-type Password]

Profile Image: [Upload button]

Figure 8 User account create

References

<https://www.w3schools.com>

<https://www.tutorialspoint.com>

<https://stackoverflow.com>

<https://startbootstrap.com>